

Mini-Briefing

Pattern Matching and Machine Learning for Audio Signal Processing

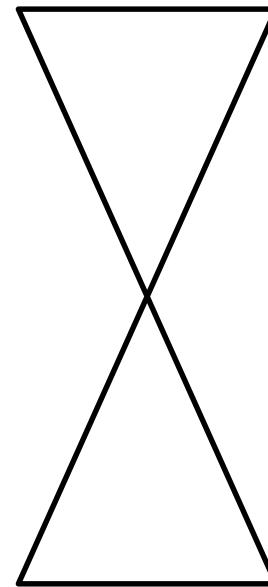
MA-INF 2212

#2 Butterfly Network

Prof. Dr. Frank Kurth



- Network structure implementing Fast Fourier Transform (FFT) algorithm
- Signal lengths $N = 2^n$
- Name „Butterfly“ due to shape of Network structure:



Task: Compute $X := \text{DFT}_N x$

Theorem 9. Let $N = 2M$ and $x \in \mathbb{C}^N$. If $x^0 = (x_0, x_2, \dots, x_{N-2})^\top \in \mathbb{C}^M$ and $x^1 = (x_1, x_3, \dots, x_{N-1})^\top \in \mathbb{C}^M$ denote the even and odd parts of x with corresponding spectra X^0 and X^1 , and if $\Delta_M := \text{diag}(1, \omega^1, \dots, \omega^{M-1})$ denotes the diagonal matrix of twiddle factors (resulting from multiplications by z), then

$$\boxed{(X(0), \dots, X(M-1))^\top = X^0 + \Delta_M \cdot X^1} \quad (\text{upper half of } X)$$

$$\boxed{(X(M), \dots, X(N-1))^\top = X^0 - \Delta_M \cdot X^1} \quad (\text{lower half of } X).$$

Comment: Decimation in time refers to splitting (decimating) the time sequence into two parts of half length – the odd and the even part.

Butterfly stage 1

$$(X(0), \dots, X(M-1))^T = X^0 + \Delta_M \cdot X^1 \quad (\text{upper half of } X)$$

$$(X(M), \dots, X(N-1))^T = X^0 - \Delta_M \cdot X^1 \quad (\text{lower half of } X).$$

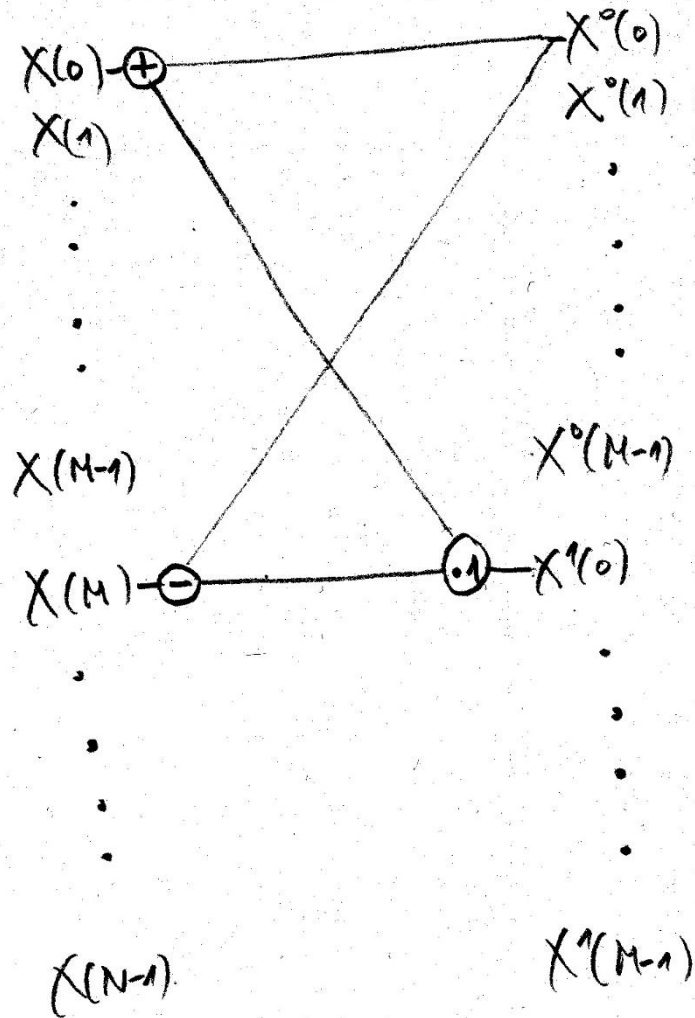
$N = 2 \cdot M$

$$\text{DFT}_N \cdot x = \begin{bmatrix} X(0) \\ \vdots \\ X(M-1) \\ X(M) \\ \vdots \\ X(N-1) \end{bmatrix} = \begin{bmatrix} X^0(0) \\ \vdots \\ X^0(M-1) \\ X^0(0) \\ \vdots \\ X^0(M-1) \end{bmatrix} + \begin{bmatrix} 1 & & \\ & \ddots & \\ & & \omega^{M-1} \end{bmatrix} \begin{bmatrix} X^1(0) \\ \vdots \\ X^1(M-1) \end{bmatrix} - \begin{bmatrix} 1 & & \\ & \ddots & \\ & & \omega^{M-1} \end{bmatrix} \begin{bmatrix} X^1(0) \\ \vdots \\ X^1(M-1) \end{bmatrix}$$

$= \Delta_M$

$$= \begin{bmatrix} \text{DFT}_M \cdot X^0 & + \Delta_M \cdot \text{DFT}_M \cdot X^1 \\ \text{DFT}_M \cdot X^0 & - \Delta_M \cdot \text{DFT}_M \cdot X^1 \end{bmatrix}$$

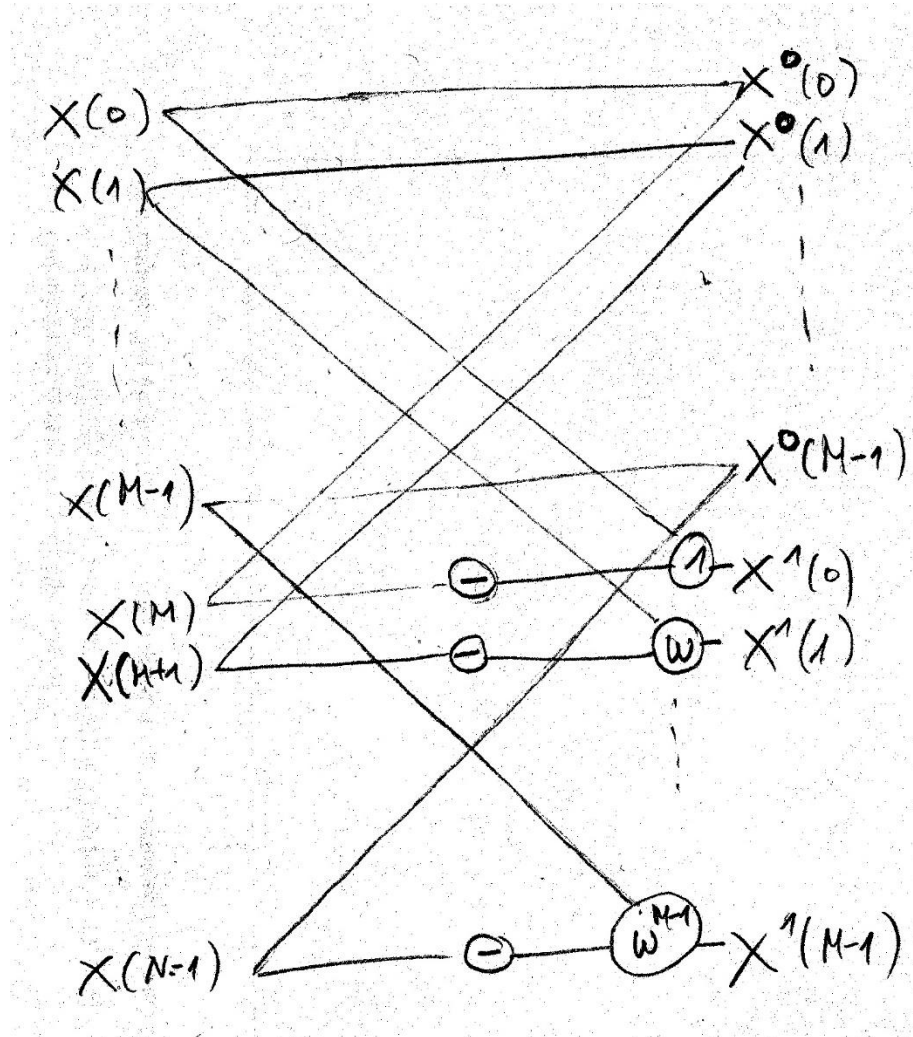
Butterfly stage 1



$$\begin{bmatrix} X^o(0) \\ \vdots \\ X^o(M-1) \end{bmatrix} + \begin{bmatrix} 1 & & \\ & \ddots & \\ & & w^{M-1} \end{bmatrix} \begin{bmatrix} X^o(0) \\ \vdots \\ X^o(M-1) \end{bmatrix} = \begin{bmatrix} X^{\wedge}(0) \\ \vdots \\ X^{\wedge}(M-1) \end{bmatrix}$$

$$\begin{bmatrix} X^o(0) \\ \vdots \\ X^o(M-1) \end{bmatrix} - \begin{bmatrix} 1 & & \\ & \ddots & \\ & & w^{M-1} \end{bmatrix} \begin{bmatrix} X^o(0) \\ \vdots \\ X^o(M-1) \end{bmatrix} = \begin{bmatrix} X^{\wedge}(0) \\ \vdots \\ X^{\wedge}(M-1) \end{bmatrix}$$

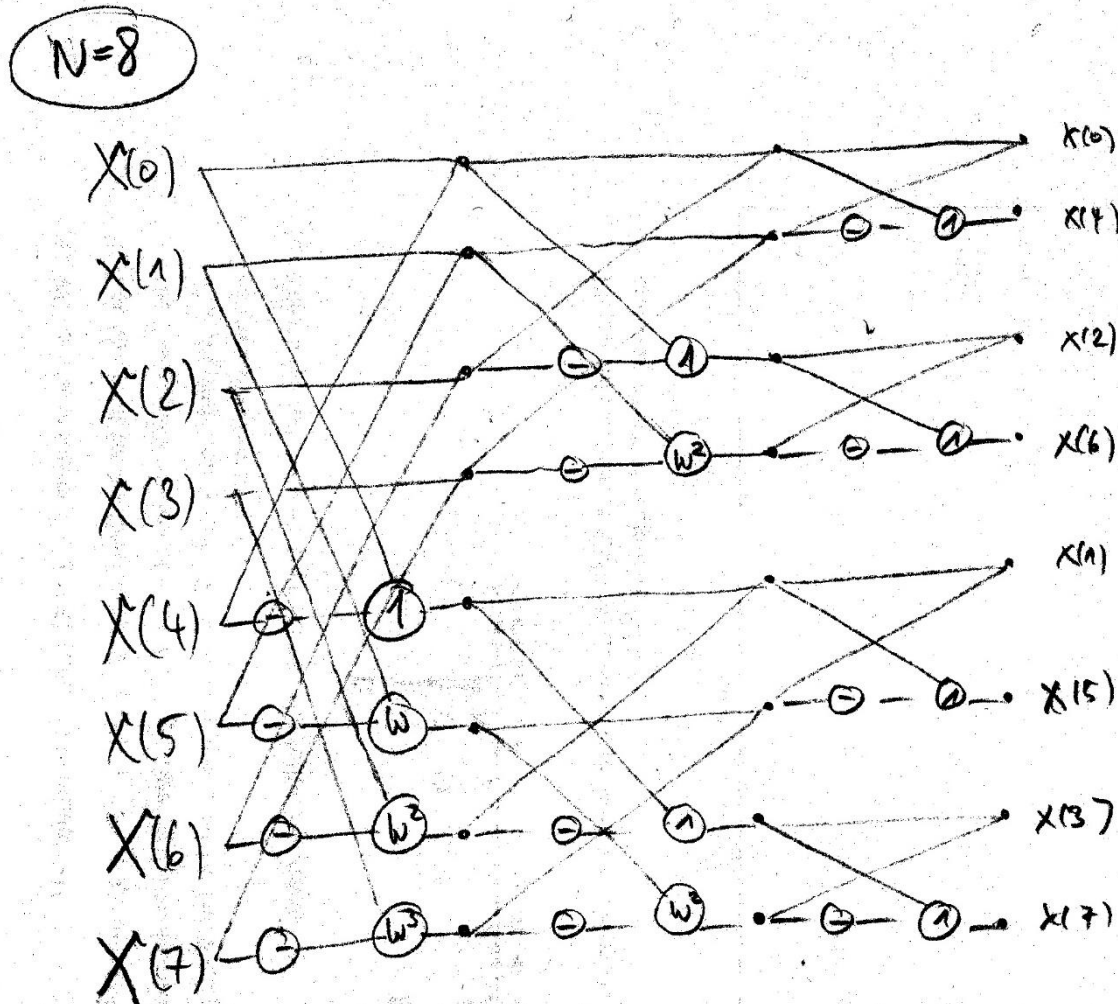
Butterfly stage 1



$$\begin{bmatrix} X^o(0) \\ \vdots \\ X^o(M-1) \end{bmatrix} + \begin{bmatrix} 1 & & \\ & \ddots & \\ & & w^{M-1} \end{bmatrix} \begin{bmatrix} X^1(0) \\ \vdots \\ X^1(M-1) \end{bmatrix}$$

$$\begin{bmatrix} X^o(0) \\ \vdots \\ X^o(M-1) \end{bmatrix} - \begin{bmatrix} 1 & & \\ & \ddots & \\ & & w^{M-1} \end{bmatrix} \begin{bmatrix} X^1(0) \\ \vdots \\ X^1(M-1) \end{bmatrix}$$

Full butterfly network



$$w = w_8 = e^{-\frac{2\pi i}{8}}, \quad w^0 = 1$$

- Also: Decimation in frequency FFT
- Efficient FFTs, particularly for signal lengths $N = 2^n, 4^n, 8^n$
 - Radix-2-FFT, Radix-4-FFT, Radix-8-FFT
- Efficient FFTs for prime factor lengths p



- Mixed radix DFT:
 - Let $N = p \cdot q$,
 - DFT_N can be computed by DFT_p 's and DFT_q 's

Example: $N = M \cdot 2$, X^0 and X^1 computed via DFT_M

$$(X(0), \dots, X(M-1))^T = X^0 + \Delta_M \cdot X^1 \quad (\text{upper half of } X)$$

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FFTs – Remarks (2)

- Mixed radix DFT:
 - Let $N = p \cdot q$,
 - DFT_N can be computed by DFT_p 's and DFT_q 's

Example: $N = M \cdot 2$, X^0 and X^1 computed via DFT_M

$$(X(0), \dots, X(M-1))^T = \underset{1}{X^0} + \Delta_M \cdot \underset{1}{X^1} \quad (\text{upper half of } X)$$

$$(X(M), \dots, X(N-1))^T = \underset{1}{X^0} - \Delta_M \cdot \underset{-1}{X^1} \quad (\text{lower half of } X).$$

DFT_2 hidden here

